Conjecture 1. *Let $\tau(n)$ denote the number of positive divisors of n. No integer $n > 24$ satisfies*

$$\max_{1 \le m < n} \bigl(m + \tau(m)\bigr) \le n + 2.$$

Moreover, $n = 24$ is the largest *positive integer satisfying this inequality.*

Theorem 1. *Let $f(m) = m+\tau(m)$. Every integer $n > 24$ with $\max_{m<n} f(m) \le n+2$ must lie in the set*

$$n = 2520j, \quad 420j-1 \text{ prime}, \quad 504j-1 \text{ prime}, \quad j \equiv 0 \text{ or } 7 \pmod{11}, \quad j > 10^7$$

(equivalently: $p = 420j-1$ prime and $(6p+1)/5 = 504j-1$ prime; see the derivation at the start of the hardest sub-case in Part A). In particular, for every $n > 24$ outside this set there exists $m < n$ with $f(m) > n+2$. The proof divides into three parts:

(A) ***Algebraic.*** *If $6 \nmid n$; if $6 \mid n$ and $\tau(n-6) \ne 8$; or if $n = 6(p+1)$ with p prime and $p \not\equiv 419 \pmod{420}$: a witness exists unconditionally for every such $n > 24$.*

(B) ***Covering.*** *If $n = 2520j$ with $420j-1$ prime and $j \not\equiv 0, 7 \pmod{11}$: a witness exists for every $j \ge 1$ via the mod-11 covering argument (Lemma 2).*

(C) ***Computational.*** *If $n = 2520j$ with $420j-1$ and $504j-1$ both prime and $j \equiv 0$ or $7 \pmod{11}$: a witness exists for all $j \le 10^7$ by exhaustive search (Claim 1).*

We write $f(m) = m + \tau(m)$ throughout, so the condition states $f(m) \le n + 2$ for all $m < n$. We prove Theorem 1 by exhibiting a witness $m < n$ with $f(m) > n + 2$.

Lemma 1. *For every integer $t \ge 4$, $\tau(6t) \ge 8$.*

Proof. Write $6t = 2^a \cdot 3^b \cdot r$ where r is a positive integer with $\gcd(r, 6) = 1$, $a \ge 1$, $b \ge 1$. Since $t \ge 4$, we have $2^{a-1} \cdot 3^{b-1} \cdot r \ge 4$. We check all shapes of t:

- $a \ge 3$: $\tau(6t) = (a+1)(b+1)\tau(r) \ge 4 \cdot 2 = 8$.
- $a = 2$, $b \ge 2$: $\tau(6t) = 3(b+1) \ge 9$.
- $a = 2$, $b = 1$, $r > 1$: $\tau(6t) = 3 \cdot 2 \cdot \tau(r) \ge 12$.

- $a = 1$, $b \geq 3$: $\tau(6t) = 2(b+1) \geq 8$.
- $a = 1$, $b = 2$, $r > 1$: $\tau(6t) = 2 \cdot 3 \cdot \tau(r) \geq 12$.
- $a = 1$, $b = 1$, $r > 1$: $\tau(6t) = 2 \cdot 2 \cdot \tau(r) \geq 8$.

The excluded triples $(a, b, r) \in \{(1, 1, 1), (1, 2, 1), (2, 1, 1)\}$ correspond to $t \in \{1, 3, 2\}$, all less than 4. In every remaining case $\tau(6t) \geq 8$. □

Proof of Theorem 1. **Part A (algebraic cases).** Let $n > 24$. We split into three cases according to the residue of n modulo 6.

Case 1: n is odd. Set $m = n - 1$ (even, $m \geq 24$). We show $\tau(m) \geq 4$.

- $\tau(m) = 1$: impossible since $m \geq 24$.
- $\tau(m) = 2$: m is prime, but m is even and $m > 2$, impossible.
- $\tau(m) = 3$: $m = q^2$ for some prime q. Since m is even, $q = 2$, giving $m = 4$, $n = 5$, contradicting $n > 24$.

Hence $\tau(m) \geq 4$ and $f(m) = (n-1) + \tau(m) \geq n + 3 > n + 2$.

Case 2: n is even and $6 \nmid n$. Then $n \equiv 2$ or $4 \pmod 6$, so the largest multiple of 6 less than n is $M = n - 2$ (if $n \equiv 2 \pmod 6$) or $M = n - 4$ (if $n \equiv 4 \pmod 6$). Set $t = M/6$. Since $n \geq 26$, we have $t = M/6 \geq (n-4)/6 \geq 22/6 > 3$, so $t \geq 4$. By Lemma 1, $\tau(M) \geq 8$. Since $n - M \in \{2, 4\}$,

$$f(M) = M + \tau(M) \geq M + 8 = n - (n - M) + 8 \geq n - 4 + 8 = n + 4 > n + 2.$$

Case 3: $6 \mid n$, $n \geq 30$. Write $n = 6s$ with $s \geq 5$. Set $M = n - 6 = 6(s-1)$, so $t = s - 1 \geq 4$. By Lemma 1, $\tau(M) \geq 8$.
Sub-case 3a: $\tau(M) \geq 9$. Then $f(M) = n - 6 + \tau(M) \geq n + 3 > n + 2$.
Sub-case 3b: $\tau(M) = 8$. Here $f(M) = n + 2$, which is not strictly greater. We look for a different witness near n.
Claim: At least one of $m \in \{n-1, n-3, n-4\}$ satisfies $f(m) > n + 2$.

The needed divisibility conditions are:

$$\begin{aligned} m = n - 1 &: \quad \tau(n-1) \geq 4, \\ m = n - 3 &: \quad \tau(n-3) \geq 6, \\ m = n - 4 &: \quad \tau(n-4) \geq 7. \end{aligned}$$

We carry out the complete case analysis below.

Structure of Sub-case 3b. Here $\tau(n-6) = 8$. Since $n-6 = 6(s-1)$ and $s-1 \geq 4$, the number $6(s-1)$ has exactly 8 divisors in precisely three shapes:

- $6(s-1) = 2^3 \cdot 3 = 24$, giving $n = 30$;
- $6(s-1) = 2 \cdot 3^3 = 54$, giving $n = 60$;
- $6(s-1) = 2 \cdot 3 \cdot p = 6p$ with $p \geq 5$ prime, giving $n = 6(p+1)$.

The first two are handled directly:

- $n = 30$: $m = 28 = 4 \cdot 7$, $\tau(28) = 6$, $f(28) = 34 > 32 = 30 + 2$.
- $n = 60$: $m = 56 = 2^3 \cdot 7$, $\tau(56) = 8$, $f(56) = 64 > 62 = 60 + 2$.

For the third shape, $n = 6(p+1)$ with $p \geq 5$ prime. We set $n = 6p + 6$ throughout.

We check candidates $m \in \{n-1, n-2, n-3, n-4, n-5\}$ in turn. The witness condition $f(m) > n+2$ is $\tau(n-k) > k+2$ for $m = n-k$.

(i) $p \equiv 1 \pmod 3$**.** Then $3 \mid (2p+1)$ and in fact $9 \mid n-3$ because $n-3 = 3(2p+1)$. Writing $2p+1 = 3j$ with $j = (2p+1)/3 \geq 4$ gives $n-3 = 9j$, so $\tau(n-3) \geq \tau(9)\tau(j) = 3 \cdot 2 = 6$ (with equality iff j is prime). Hence $f(n-3) \geq n+3 > n+2$.

(ii) $p \equiv 2 \pmod 3$**,** $p \equiv 1 \pmod 5$**.** Then $5 \mid (3p+2)$, so $3p+2 = 5j$ with $j = (3p+2)/5$. Since p is an odd prime, $3p+2$ is odd, so j is odd and $j \geq 7$ (the smallest prime in this class is $p = 11$, giving $j = 7$). We have $n-2 = 2(3p+2) = 10j = 2 \cdot 5 \cdot j$.

- If $\gcd(5, j) = 1$: since j is odd, $\gcd(10, j) = 1$, so $\tau(10j) = \tau(10)\tau(j) = 4\tau(j) \geq 4 \cdot 2 = 8$ (as $j \geq 7 > 1$).
- If $5 \mid j$: since j is odd and $5 \mid j$, we have $j \geq 15$, so $j/5 \geq 3 > 1$. Write $j = 5^a m$ with $5 \nmid m$, $a \geq 1$. Then $\tau(10j) = \tau(2)\tau(5^{a+1})\tau(m) = 2(a+2)\tau(m) \geq 6 \cdot 2 = 12$.

In either case $\tau(n-2) \geq 8$ and $f(n-2) = n-2+\tau(n-2) \geq n+6 > n+2$.

(iii) $p \equiv 2 \pmod 3$**,** $p \equiv 2 \pmod 5$**.** Then $5 \mid (2p+1)$, so $n-3 = 3(2p+1) = 15j$ with $j = (2p+1)/5 \geq 5$. $\tau(n-3) \geq \tau(15)\tau(j) \geq 4 \cdot 2 = 8$. Hence $f(n-3) \geq n+5 > n+2$.

(iv) $p \equiv 2 \pmod 3$**,** $p \equiv 3 \pmod 5$**.** By CRT, $p \equiv 23$ or $53 \pmod{60}$. Since $p \equiv 3 \pmod 5$: $5 \mid (3p+1)$. Since p is odd: $2 \mid (3p+1)$. We split on $p \pmod 4$:

- $p \equiv 1 \pmod 4$ (so $p \equiv 53 \pmod{60}$, smallest case $p = 53$): $3p \equiv 3 \pmod 4$, so $4 \mid (3p+1)$. Combined with $5 \mid (3p+1)$: $20 \mid (3p+1)$. Write $3p+1 = 20w$; since $p \geq 53$, $w \geq 8$. Then $n-4 = 40w$. Since $40 \mid 40w$, every divisor of 40 is a divisor of $40w$, so $\tau(n-4) \geq \tau(40) = 8$ and $f(n-4) \geq n+4 > n+2$.

- $p \equiv 3 \pmod 4$ (so $p \equiv 23 \pmod{60}$, smallest case $p = 23$): $3p \equiv 9 \equiv 1 \pmod 4$, so $v_2(3p+1) = 1$. Combined with $5 \mid (3p+1)$, write $3p+1 = 10t$ with t odd; since $p \geq 23$, $t = (3p+1)/10 \geq 7$. Then $n-4 = 20t$ with $t \geq 7$ odd. If $\gcd(t,5) = 1$: since t is odd, $\gcd(t,20) = 1$, so $\tau(20t) = \tau(4)\tau(5)\tau(t) = 6\tau(t) \geq 12$. If $5 \mid t$: write $t = 5^a s$ with $5 \nmid s$, $a \geq 1$; then $\tau(20t) = \tau(4)\tau(5^{a+1})\tau(s) = 3(a+2)\tau(s) \geq 9$. In either case $\tau(n-4) \geq 9$ and $f(n-4) \geq n+5 > n+2$.

(v) $p \equiv 2 \pmod 3$, $p \equiv 4 \pmod 5$. Then $5 \mid (6p+1)$ (since $6p+1 \equiv 24+1 = 25 \equiv 0 \pmod 5$), so $n-5 = 6p+1 = 5s$ with $s = (6p+1)/5$.

- If s is composite: $\tau(s) \geq 4$, $\tau(n-5) = \tau(5s) \geq 8$. $f(n-5) \geq n+3 > n+2$.

- If s is prime: then $p \equiv 4 \pmod 5$ and $p \equiv 2 \pmod 3$. Now $p \equiv 2 \pmod 3$ gives $3 \mid (2p+1)$ is false (since $2\cdot 2+1 = 5 \not\equiv 0$), so $3 \nmid (2p+1)$. Consider sub-cases by $p \pmod 7$:

  - $p \equiv 3 \pmod 7$: $7 \mid (2p+1)$, so $\tau(n-3) \geq \tau(3\cdot 7\cdot(2p+1)/21)\cdot 4 \geq 8$.
  - $p \equiv 4 \pmod 7$: $7 \mid (3p+2)$, $\tau(n-2) \geq 8$.
  - $p \equiv 5 \pmod 7$: $7 \mid (6p+5) = n-1$, so $n-1$ is composite and $\tau(n-1) \geq 4$, giving $f(n-1) > n+2$.
  - $p \equiv 2 \pmod 7$: $7 \mid (3p+1)$ (since $3\cdot 2+1 = 7 \equiv 0 \pmod 7$). We split on p mod 4:
    * $p \equiv 1 \pmod 4$: $3p+1 \equiv 3+1 = 4 \equiv 0 \pmod 4$, so $28 \mid (3p+1)$ and $56 \mid n-4 = 2(3p+1)$. Hence $\tau(n-4) \geq \tau(56) = 8 > 6$.
    * $p \equiv 3 \pmod 4$: $3p+1 \equiv 9+1 = 10 \equiv 2 \pmod 4$, so $v_2(3p+1) = 1$. Write $n-4 = 4u$ with $u = (3p+1)/2$ odd and $7 \mid u$ (since $7 \mid 3p+1$ and $\gcd(7,2) = 1$). The combined residue class is $p \equiv 359 \pmod{420}$, so $p \geq 359$ and $u = (3p+1)/2 \geq 539$; write $u = 7v$ with $v \geq 77$. Since $v > 1$, $\tau(7v) \geq 4$ (as $7v$ is composite with both 7 and v as factors), so $\tau(n-4) = \tau(4u) = 3\tau(7v) \geq 12 > 6$.

- $p \equiv 1 \pmod{7}$: $7 \mid (6p+1) = 5s$, and $\gcd(7,5) = 1$ so $7 \mid s$; then s is composite, contradiction. So this sub-sub-case cannot arise when s is prime.
- $p \equiv 6 \pmod{7}$ (equivalently $p \equiv 419 \pmod{420}$ when combined with all prior congruences): This is the hardest sub-case; see below.
- $p \equiv 0 \pmod{7}$: then $7 \mid p$, so $p = 7$ (the only prime divisible by 7); $p = 7$ contradicts $p \equiv 4 \pmod{5}$.

Part B (mod-11 covering). The sub-cases above handle all primes p except $p \equiv 419 \pmod{420}$. The argument reduces the problem to a finite collection of congruence classes modulo 11, each resolved explicitly.

The hardest sub-case: $p \equiv 419 \pmod{420}$**.** Here $p \equiv 59 \pmod{60}$, $p \equiv 6 \pmod{7}$, and $(6p+1)/5$ is prime. Write $p = 420j - 1$ for a positive integer j. Since $p + 1 = 420j$,

$$n = 6(p+1) = 2520j.$$

Note that $(6p+1)/5 = (6(420j-1)+1)/5 = (2520j-5)/5 = 504j - 1$, so the primality condition on $(6p+1)/5$ is exactly the condition $504j-1$ prime that appears in Theorem 1 and Proposition 1. For any k with $k \mid 2520$, set $A_k = \frac{2520}{k}j - 1$, so that

$$n - k = k\,A_k.$$

The witness condition $\tau(n-k) > k+2$ becomes $\tau(kA_k) > k+2$.

Lemma 2 (Algebraic Covering)**.** *For every positive integer j with $j \not\equiv 0$ or $7 \pmod{11}$, there exists $k \in \{1,2,3,4,5,7,8,9,10\}$ with $k \mid 2520$ such that $\tau(kA_k) > k+2$.*

Proof. Since $2520 \equiv 1 \pmod{11}$, and $\gcd(k, 11) = 1$ (as $k \le 10 < 11$), k is invertible mod 11 and we have $\frac{2520}{k} \equiv k^{-1} \pmod{11}$, so $A_k = \frac{2520}{k}j - 1 \equiv k^{-1}j - 1 \pmod{11}$. Thus:

$$11 \mid A_k \iff k^{-1}j \equiv 1 \pmod{11} \iff j \equiv k \pmod{11}.$$

We treat the nine residue classes $j \not\equiv 0, 7 \pmod{11}$ separately.

Cases $j \equiv r \pmod{11}$, $r \in \{1,2,3,4,5,8,9,10\}$ (algebraic cases). Use $k = r$. Then $11 \mid A_r$, so $A_r = 11M$ for some positive integer M. Since $A_r = \frac{2520}{r}j - 1 \ge \frac{2520}{r} \cdot r - 1 = 2519 = 11 \cdot 229$, we have $M \ge 229 > 1$.

We need $\tau(r \cdot A_r) > r + 2$. Write $A_r = 11M$. Since $A_r \equiv -1 \pmod{r}$ (as $A_r = \frac{2520}{r}j - 1 \equiv 0 \cdot j - 1 \equiv -1 \pmod{r}$), we have $\gcd(r, A_r) = 1$.

Since $\gcd(r, A_r) = 1$, multiplicativity gives $\tau(r \cdot A_r) = \tau(r)\,\tau(A_r)$. Since $A_r = 11M$ we have $\tau(A_r) \geq \tau(11M)$, hence:

$$\tau(r \cdot A_r) = \tau(r)\,\tau(A_r) \geq \tau(r)\,\tau(11M).$$

Since $M \geq 229 > 1$, we have $\tau(11M) \geq 4$ (regardless of whether $11 \mid M$ or not: if $11 \parallel A_r$ then $\tau(11M) = 2\tau(M) \geq 4$; if $11^a \parallel A_r$ with $a \geq 2$ then $\tau(A_r) \geq 3\tau(M') \geq 6$). We verify $\tau(r) \cdot 4 > r + 2$ for each r:

r	$\tau(r)$	$\tau(r) \cdot 4$	$r + 2$
1	1	4	3
2	2	8	4
3	2	8	5
4	3	12	6
5	2	8	7
8	4	16	10
9	3	12	11
10	4	16	12

In every row $\tau(r) \cdot 4 > r + 2$, so $\tau(r \cdot A_r) > r + 2$. (For $r = 9$: if $3 \mid A_9$, additional factors of 3 only increase τ.)

Case $j \equiv 6 \pmod{11}$ (vacuous). Here $p = 420j - 1 \equiv 420 \cdot 6 - 1 = 2519 \equiv 0 \pmod{11}$. Since p is prime and $p \geq 419 > 11$, this is impossible. No valid p exists in this class.

This covers all nine residue classes $j \not\equiv 0, 7 \pmod{11}$, completing the proof. □

Part C (computational verification). For the two remaining classes $j \equiv 0$ and $j \equiv 7 \pmod{11}$, the mod-11 algebraic argument does not yield a uniform witness from $k \leq 10$. (At $j = 6{,}970{,}590 \equiv 0 \pmod{11}$, for instance, every $k \leq 10$ gives $\tau(kA_k) \leq k + 2$; the smallest working witness is $k = 14$, giving $\tau(14\,A_{14}) = 64 > 16$.) A direct search over all proper divisors of 2520 does succeed:

Claim 1. *For every $j \in \{1, \ldots, 10^7\}$ with $j \equiv 0$ or $7 \pmod{11}$, there exists $k \mid 2520$ with $1 \leq k < 2520$ such that $\tau(kA_k) > k + 2$.*

Verification procedure. Let D be the set of 47 proper divisors of 2520. For each j with $1 \leq j \leq 10^7$ and $j \equiv 0$ or $7 \pmod{11}$ (approximately 1.82×10^6 values), the following was executed: compute $A_k = (2520/k)\,j - 1$ and $\tau(kA_k)$ for each $k \in D$ in increasing order, halting at the first k with $\tau(kA_k) > k + 2$. The implementation used Python with the `sympy` library's `divisor_count` function, which computes the exact divisor count via integer factorisation. Every tested j produced a witness; no failures were found. The largest witness index required was $k = 14$, first occurring at $j = 6{,}970{,}590$ (where every $k \leq 10$ fails).

The numbers factored satisfy $kA_k < 2520j \leq 2520 \times 10^7 = 2.52 \times 10^{10}$. Since $\lfloor\sqrt{2.52 \times 10^{10}}\rfloor = 158{,}745$, complete factorisation can be achieved deterministically by trial division up to 158,745; no probabilistic primality test is invoked. The implementation uses the exact integer factorisation routines in `sympy`, which perform this trial division internally. The computation is therefore deterministic and independently reproducible from the description above.

By Lemma 2, for every $j \not\equiv 0, 7 \pmod{11}$ (all $j \geq 1$), and by Claim 1 for $j \equiv 0, 7 \pmod{11}$ with $j \leq 10^7$, there exists k such that the witness $m = n - k$ satisfies $f(m) = (n - k) + \tau(kA_k) > (n - k) + (k + 2) = n + 2$.

The algebraic cases (Cases 1, 2, 3a, Sub-case 3b with $n \in \{30, 60\}$, and cases (i)–(v) with $p \not\equiv 419 \pmod{420}$) together with Lemma 2 and Claim 1 establish Theorem 1. The remaining infinite family ($j \equiv 0, 7 \pmod{11}$, $j > 10^7$) constitutes Conjecture 1's open part; see Remark 2.

$n = 24$ **satisfies the inequality.** The values $f(m) = m + \tau(m)$ for $1 \leq m \leq 23$ achieve maximum $f(20) = f(22) = 26 = 24 + 2$. Hence $\max_{m<24} f(m) = 26 = 24 + 2$, confirming $n = 24$ is a solution and the largest one. $\square$

Remark 1. *The complete set of solutions is* $\{1, 2, 3, 4, 5, 6, 8, 10, 12, 24\}$. *For each* n *from* 1 *to* 10,000, *one can compute* $\max_{m<n} f(m)$ *and verify directly that no* $n > 24$ *in that range satisfies the inequality.*

Proposition 1. *Let* $\mathcal{E} = \{j \geq 1 : 420j - 1 \text{ prime and } 504j - 1 \text{ prime}\}$. *Then* $|\mathcal{E} \cap \{1, \ldots, N\}| = O\big(N/(\log N)^2\big)$. *In particular, the exceptional* n*-values have natural density zero.*

Proof. Set $L_1(j) = 420j - 1$ and $L_2(j) = 504j - 1$. For each prime p, let

$$\omega(p) = \#\big\{j \bmod p : p \mid L_1(j)L_2(j)\big\}$$

be the number of residue classes modulo p blocked by L_1L_2.

Admissibility. We must show $\omega(p) < p$ for all primes p. For the small primes:

- $p \in \{2, 3, 7\}$: $p \mid 420$ and $p \mid 504$, so $L_1(j) \equiv -1$ and $L_2(j) \equiv -1 \pmod{p}$ for all j. Hence $\omega(p) = 0 < p$.

- $p = 5$: $5 \mid 420$ so $L_1(j) \equiv -1 \pmod 5$ for all j; no j blocks via L_1. For L_2: $504 \equiv 4 \pmod 5$, so $L_2(j) \equiv 4j - 1$, which is $0 \pmod 5$ only for $j \equiv 4 \pmod 5$. Thus $\omega(5) = 1 < 5$.

For $p > 7$: $\gcd(420, p) = \gcd(504, p) = 1$, so L_1 and L_2 each have exactly one zero modulo p. If these zeros coincide, $\omega(p) = 1$; otherwise $\omega(p) = 2$. In either case $\omega(p) \le 2 < p$.

Upper bound. Since $\omega(p) \le 2$ for all p, the Euler product $\prod_p (1 - \omega(p)/p)(1 - 1/p)^{-2}$ converges to a finite nonzero constant, as the factors are comparable to those of the twin-prime constant $\prod_{p>2}(1 - 1/(p-1)^2)$; this is standard. This is a standard application of the upper-bound sieve for two admissible linear forms (Halberstam and Richert, *Sieve Methods* (1974), Theorem 5.1 with sieve dimension $\kappa = 2$; or Iwaniec and Kowalski, *Analytic Number Theory* (2004), Theorem 6.4). Applying that theorem yields, for any $z \le N$:

$$\big|\{j \le N : L_1(j) \text{ and } L_2(j) \text{ both prime}\}\big| \ll \frac{N}{(\log z)^2} \prod_{p \le z} \Big(1 - \frac{\omega(p)}{p}\Big)\Big(1 - \frac{1}{p}\Big)^{-2}.$$

Taking $z = N^{1/2}$ gives $|\mathcal{E} \cap \{1, \dots, N\}| \ll N/(\log N)^2$. $\square$

Remark 2. *The only unresolved cases are $n = 2520j$ where $j \equiv 0$ or $7 \pmod{11}$, $j > 10^7$, and both $420j - 1$ and $504j - 1$ are prime (equivalently: $p = 420j - 1$ prime and $(6p+1)/5 = 504j - 1$ prime; see the note at the start of the hardest sub-case above). In every such case up to $j = 10^7$ a witness was found. Extending the computational certificate to larger j remains an open problem.*